MIL-L-48288 (PA)
16 January 1974

MILITARY SPECIFICATION

LEAD ASSEMBLY
(FOR MINE, M56 SUBSYSTEM)
PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

This Specification is not approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing of one type of lead assembly designated as Lead Assembly (for Mine, M56 SubSystem).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Method of
MIL-R-398 - RDX
MIL-A-2550 - Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for

FSC: 1345

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MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-650 - Explosives; Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

PRODUCT AND PACKAGING DRAWINGS

7548103 - Box, Set-up, Packing, Ammunition, Outer
7548104 - Box, Set-up, Packing, Ammunition, Inner
9204229 - Box, Packing, Ammunition
9287609 - Lead Assembly

INSPECTION EQUIPMENT DRAWINGS

9278644 - Firing Assembly and Details

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Lead Assembly. - The lead assembly shall comply with all requirements specified on Drawing 9287609 and with all requirements specified in the applicable specifications.

3.3 Moisture Content. - The moisture content of the RDX composition at the time and site of loading shall not exceed 0.20 percent when measured as specified in test method 101.5 of MIL-STD-650.

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3.4 Functioning. - The Lead Assembly shall be capable of high order initiation by a T20E1 Electric Detonator across a minimum air gap of 0.150 inches. High order detonation will be evidenced by a depression of a minimum depth of 0.0135 inches in the steel witness block specified in test 301.1 of MIL-STD-331.

3.5 Workmanship. - All parts and assemblies shall be fabricated, loaded and assembled in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.6 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

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with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of 150 loaded lead assemblies, 10 loaded lead assemblies (prior to insertion of closure disc and crimping), and 50 complete sets of parts (50 each of every component) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all of the exam-

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inations and tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. - If any component or subassembly fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an inert component or subassembly to comply with any of the stated requirements.

4.2.4 Loaded Lead Assemblies. - A sample of one hundred sixty (150) loaded lead assemblies and 10 loaded lead assemblies (prior to insertion of closure disc and crimping) shall be submitted in accordance with the provisions of 4.2.1 for testing as specified in 4.4.3 and 4.4.4.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accor-

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dance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup, lead (see dwg. 9287697 covering a detail of dwg. 9287609).

Categories	defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Length	Gage	01001
102.	Outside diameter, max.	Gage	01002
103.	Thickness of bottom	Gage	01003
104.	True position, inside diameter with outside diameter ...	Gage	01004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	01005

4.3.2.2 Cover, Lead (see dwg. 9287698 covering a detail of dwg. 9287609).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Thickness	Gage	02001
102.	Diameter, min.	Gage	02002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	02003

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4.2.2.3 Assembly (see dwg. 9287609).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cracks, splits, or cuts	Visual	03001
Major:	AQL 0.65 percent		
101.	Total length	Gage	03002
102.	Outside diameter, max.	Gage	03003
103.	Lacquer seal missing or color incorrect	Visual	03004
104.	Assembly distorted or damaged to extent that function may be impaired	Visual	03005
Minor:	AQL 1.50 percent		
201.	Crimp not full 360 degrees ...	Visual	03006
202.	Disc folded or wrinkled	Visual	03007
203.	Evidence of poor workmanship other than critical above (see 3.5)	Visual	03008

4.3.2.4 Box, set-up, packing, ammunition, inner (see dwg. 7548104).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Box damaged exposing contents ..	Visual	04001
102.	Label missing or incorrect	Visual	04002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	04003

4.3.2.5 Box, set-up, packing, ammunition, outer (see dwg. 7548103).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major:		AQL 0.40 percent	
101.	Barrier bag damaged	Visual	05001
102.	Label missing or incorrect	Visual	05002

Minor:		AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.5)	Visual	05003

4.3.2.6 Box, Packing, Ammunition (see dwg. 9204229 for Lead Cup Assembly XM804E1 and Lead Assembly for Mine, M56 Subsystem.

Categories	Defects	Method of Inspection	Code No.

Critical: None defined

Major:		AQL 0.65 percent	
101.	Box damaged	Visual	06001
102.	Nailing inadequate	Visual	06002
103.	Strapping missing or loose	Visual	06003
104.	Marking incomplete, misleading or unidentifiable	Visual	06004

Minor:		AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.5)	Visual	06005

4.3.3 TESTING

4.3.3.1 Moisture Content (see 3.3). - Major defect - Code No. 03009. - The contractor shall provide adequate controls to insure that the RDX complies with the requirements for verification, the contractor shall select a sample for test from the quantity of RDX used in each eight hours of loading production. A composite sample shall not be used. If the moisture content of the sample exceeds the requirements the loading has not begun, that quantity of RDX represented by the sample shall be rejected. If components have been loaded with RDX containing excessive moisture, the remaining unloaded RDX and the components shall be rejected. The test procedure shall be as specified in 4.4.1.

4.3.3.2 Composition (see dwg. 9287609) - Major Defect - Code No. 03010. - A sample of approximately four (4) ounces of

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unconsolidated RDX shall be selected for the performance of these analysis. Failure to comply with all the specified requirements shall be cause for rejection of the lot. The test shall be performed as specified in 4.4.2.

4.3.3.3 Functioning (see 3.4) - Major Defect - Code No. 03011. - The sample size and acceptance or rejection of the lot shall be in accordance with Table B1, Code letter P of MIL-STD-414, using an AQL of 0.15. The test shall be performed as specified in 4.4.4 using equipment specified in 4.3.4. A lead that fails to fire or fails to create a depression in the steel witness block of the required depth, when measured as outlined in MIL-STD-331, test method 301.1, shall be classed defective.

4.3.3.4 Density of explosive charge (see dwg. 9287609) Major Defect - Code No. 07001.-Ten (10) lead cup assemblies (prior to inserting closure disc and crimping) shall be selected from each lot for this determination. The test shall be performed as specified in 4.4.3. If any assembly fails to comply with the density requirements specified on dwg. 9287609, the lot shall be rejected.

4.3.3.5 Heat seal test of sealed bag (see dwg. 7548103) - Major Defect - Code No. 05003. - Sampling, acceptance criteria and test procedures for this test shall be as specified in Specification MIL-P-116.

4.3.4 Inspection Equipment. - Inspection equipment required to perform the inspections prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection design and approval.

4.4 Test Methods and Procedures

4.4.1 Moisture Content. - The moisture content of the RDX shall be determined in accordance with method 101.5 of MIL-STD-650.

4.4.2 Composition

4.4.2.1 Determination of calcium resinate content. - Transfer an accurately weighed portion of approximately 5 grams (gm) of the specimen to a tared medium porosity, sintered glass crucible. Extract the sample with three separate 20 milliliter (ml.) portions of chloroform which have been previously saturated with RDX, maintaining each portion in contact with the residue for approximately five minutes before applying suction. Stir occasionally

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to break up any lumps which may persist and wash down the sides of the crucible with each 20 ml. portion of chloroform to insure complete removal of the calcium resinate. Aspirate the residue until the order of chloroform is no longer detectable, place the crucible and contents in a drying oven at 100 degrees centigrade (°C.) to 105°C. for one half hour, cool in a desiccator and weigh. Repeat the above extraction, drying and weighing procedure with individual 20 ml. portions of chloroform until the loss in weight on subsequent extractions totals less than 0.003 gm. Calculate the loss in weight of the crucible plus contents to precent calcium resinate or stearic acid, as applicable, on a moisture-free basis as follows:

$$\text{Percent calcium resinate or stearic acid} = \frac{100A}{W}$$

where:
A = loss in weight of the crucible plus contents
W = weight of specimen on a mositure-free basis.

4.4.2.2 Determination of Graphite Content. - Extract the residue remaining after the calcium resinate has been removed with two (2) 20 ml. portions of dimethylformamide, maintaining each portion in contact with the residue for approximately five minutes with intermittent stirring before applying suction. Wash the crucible and contents with a minimum of three 20 ml. portions of acetone, maintaining each portion in contact with the residue for approximately one minute with continuous stirring, before applying suction and until the following qualitative test for RDX in the acetone washings is negative. Transfer the acetone filtrate to a 25 ml. beaker, evaporate to dryness in a steam bath and add several drops of a solution of chromotropic acid in normal (n) sulfuric acid. The formation of a reddish-brown color as compared to a pink color obtained by a blank prepared in an identical manner indicates the presence of RDX. Aspirate the residue until the odor of acetone is no longer detectable, place the crucible and contents in a drying oven at 100° to 105°C. for one half hour, cool in a desiccator and weigh. Calculate the weight of the residue to percentage of graphite in a moisture-free basis as follows:

$$\text{Percent graphite} = \frac{100A}{W}$$

where:
A = increase in weight of tared crucible
W = weight of specimen on a moisture-free basis

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4.4.2.3 Determination of total binder-lubricant content. - The percent total binder-lubricant content shall be determined by adding the percent calcium resinate obtained to the percent graphite, when applicable.

4.4.2.4 Determination RDX Content. - The percent RDX content shall be determined by subtracting the percent total binder-lubricant form 100 percent.

4.4.3 Density of explosive charge.-Prior to assembly of the closure disc and crimping, the lead cup assemblies shall be subjected to this determination during the following procedure:

a. Weigh empty lead cup (grams).
b. Measure bottom thickness (inches).
c. Measure inside diameter (inches).
d. Load cup
e. Weigh loaded lead cup (grams).
f. Measure height to top of charge (inches).
g. Calculate weight of charge (d-a) (grams).
h. Calculate height of charge (e-b) (inches).
i. Calculate volume of explosive charge
12.8727 (c x c) (h) (cubic centimeters)
j. Calculate density of explosive charge from $\frac{g}{i}$ (above)

(grams per cubic centimeter)

4.4.4 Functioning.-The lead shall be assembled in the test fixture with the T20E1 detonator as indicated on drawing 9278644. The detonator shall be fired by applying the energy to the detonator leads from a 0.4 min. capacitor charged to 50 volts maximum.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C. - The leads shall be preserved and packaged in accordance with dwg. 7548104.

5.1.2 Intermediate packaging. - The leads shall be packaged in accordance with dwg. 7548103.

5.2 Packing

5.2.1 Level C. - The containers shall be packed in accordance with dwg. 9204229.

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5.3 Marking

5.3.1 Unit, intermediate and exterior containers shall be marked in accordance with dwgs. 7548104, 7548103 and 9204229.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

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6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Check test for possible deterioration. - The following test is designated to prevent loading of leads which have deteriorated in the ammunition component with which it is to be loaded. The tests should be performed on leads from each lot which have been stored for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the leads are loaded does not require that the check tests

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be performed, the contracting officer should insert provisions for the performance of the check test in the contract. The Contracting Officer should arrange to have the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration.

6.4.1 Functioning. - The sample size and acceptance or rejection criteria shall be in accordance with Table B1, Code Letter P of MIL-STD-414, using an AQL of 0.40. The test shall be performed as specified in 4.3.3.3 using equipment specified in 4.3.4.

6.5 Submission of test data and ammunition data cards. - In addition to the normal distribution of records, one copy of all ammunition data cards shall be forwarded to each of the following:

Commander
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: (1) SARPA-QA-A-M
(2) SARPA-AD-E-S-2

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1345-A-040

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